Quality Information Document

Ocean Monitoring Indicator (OMI) of the Copernicus Marine Service

EXTREME SST

Sea surface temperature

Issue: 2.1

Contributors : Marta de Alfonso, Alejandro Gallardo, Fernando Manzano, Begoña Pérez Gómez

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MERCATOR OCEAN
INTERNATIONAL
Parc Technologique du Canal - 8-10 rue Hermès - 31520 Ramonville-Saint-Agne, FRANCE
Tél : +33 5 61 39 38 02 - Fax : +33 5 61 39 38 99
Société civile de droit français au capital de 2 000 000 € - 522 911 577 RCS Toulouse - SIRET 522 911 577 00016

marine.copernicus.eu
mercator-ocean.eu

CHANGE RECORD

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
	22/07/2019	All	Creation of the document	Marta de Alfonso, Alejandro Gallardo, Fernando Manzano, Begoña Pérez Gómez	
1.0	02/10/2019	All	Adapt the document to the new OMI structure and naming	Marta de Alfonso	
2.0	02/09/2022		New Product naming and new template. Change of variable temp by sst.	Marta de Alfonso	Stéphane Tarot
2.1	07/05/2025		Estimation of uncertainties with standard deviation of percentiles	Marta de Alfonso, Begoña Pérez Gómez, Alex Gallardo	Stéphane Tarot

I INTRODUCTION *4*
I.1 Product name 4
I.2 Indicator evaluation method 4
I.3 Products used as input for the computations 5
II SCIENTIFIC VALIDATION *6*
III REFERENCES *8*

I INTRODUCTION

I.1 Product name

OMI_EXTREME_SST_BALTIC_sst_mean_and_anomaly_obs

OMI_EXTREME_SST_NORTHWESTSHELF_sst_mean_and_anomaly_obs

OMI_EXTREME_SST_WAVE_IBI_sst_mean_and_anomaly_obs

OMI_EXTREME_SST_MEDSEA_sst_mean_and_anomaly_obs

I.2 Indicator evaluation method

The EXTREME SST indicators are based on the computation of the 99th and the 1st percentiles (that represent approximately the 1% highest and lowest values, respectively) from in situ data (observations), following the same approach applied by Woodworth and Blackman (2004) to extreme sea levels. This method determines changes in the frequency distribution of the measured variables.

All the data used for the computation of these indicators have been downloaded from Copernicus Marine Service In Situ Thematic Assembly Centre (In Situ TAC) products. The analysis is performed for the following variable: sea surface temperature (sst) measured by in situ buoys.

The periods selected are based on our analysis of the data available in the different In Situ TAC regions that fulfilled specific criteria of data completeness and quality (Pérez Gómez et al 2016, Pérez Gómez et al 2018). Buoys were selected according to consolidated time series of hourly data in Copernicus Marine Service for the period 2000 to the target year. Within these periods, a minimum number of 10 years with completeness index larger than 70% is required for a station to be included. These numbers can be reduced to 8 years and 60% in case that the coverage is not enough to make the OMIs properly representative. These data were then used to compute annually the 99th and 1st percentile levels at each station.

More details can be found in the Copernicus Marine Service Ocean State Report (Pérez Gómez et al., 2016; Pérez Gómez et al., 2018).

I.3 Products used as input for the computations

Product name (and Type)	Documentation
INSITU_GLO_PHYBGCWAV_DISCRETE_MYNRT_013_030 INSITU_BAL_PHYBGCWAV_DISCRETE_MYNRT_013_032 INSITU_NWS_PHYBGCWAV_DISCRETE_MYNRT_013_036 INSITU_IBI_PHYBGCWAV_DISCRETE_MYNRT_013_033 INSITU_MED_PHYBGCWAV_DISCRETE_MYNRT_013_035	QUID: https://doi.org/10.13155/75807 PUM: https://doi.org/10.13155/43494

II SCIENTIFIC VALIDATION

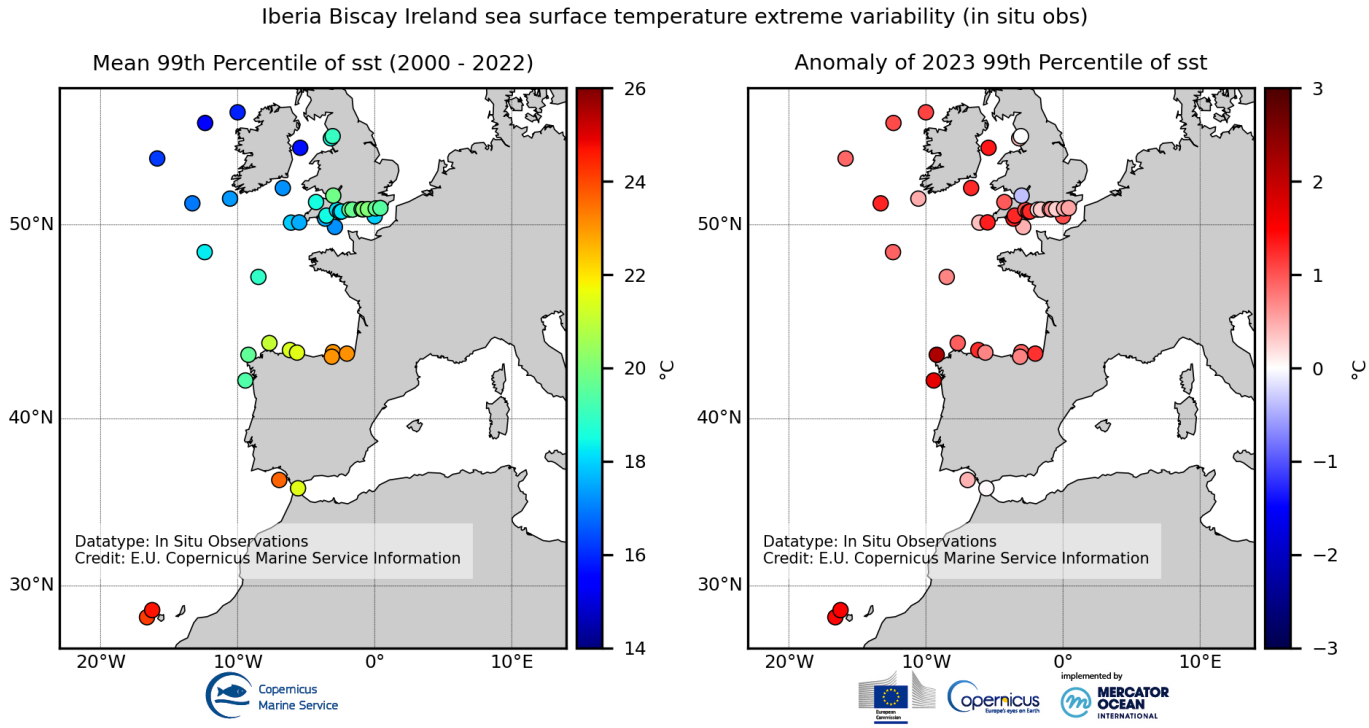
The quality validation of the EXTREME SST indicators is based on the validation of the original in situ data used to compute the percentiles.

The fixed stations used for these OMIs are not included in the In Situ TAC reprocessed products (MY), so the information has been downloaded from the In Situ TAC Near Real Time products (NRT) to perform the analysis. Accuracy numbers about the original data and the quality procedures applied to the NRT products can be found in the Quality Information Document of NRT In Situ TAC products (https://doi.org/10.13155/75807). Complete information about the quality procedures applied to In Situ TAC NRT data can be found in the Copernic Marine In Situ TAC Quality Control Procedures (https://doi.org/10.13155/106219).

For these NRT data, a visual inspection by scientists has been performed to assure that no spurious value interferes and deviate the extremes results.

In a normal year with no data gaps, we have 8760 hourly values (6132 values for the minimum data coverage of 70%). For these numbers, the 99th and 1st percentiles represent the 88 highest and lowest hourly values, respectively (or 61 for the 70% minimum data coverage). This makes this extremes study relatively immune to individual data measurement errors instead of annual maximum and minimum values.

Apart from the robustness provided using percentiles for the extremes study, we check the consistency by performing an analysis of the spatial distribution, starting with the calculation of the percentiles for every location and for every year. In the analysis (corresponding to the products: OMI_EXTREME_SST_[region]_sst_mean_and_anomaly_obs), the temporal mean percentile is calculated with the whole series until the year previous to the target year and its spatial distribution is shown together with the anomaly of the target year. This analysis allows to check the geographical coherence of the OMI (see Figure below).



Iberia Biscay Ireland sea surface temperature 99th percentile. Left: Mean of annual values (2000 - 2022). Right: Anomalies of 2023 with respect to the mean.

More details can be found in the Copernicus Marine Service Ocean State Report (Pérez Gómez et al., 2016; Pérez Gómez et al., 2018).

Uncertainty

Uncertainty refers to the error associated with estimating a parameter, such as the mean, and arises due to inherent variability within a sample. This variability is quantified by the standard deviation, a statistical measure that characterizes the dispersion of data points around the mean, thereby providing an indication of the uncertainty involved in the estimation.

The standard deviation included in this indicator defines the threshold for statistical significance, with anomalies considered significant if they deviate substantially from it.

III REFERENCES

Copernicus In Situ tac Data Management Team (2025). Copernicus Marine In Situ TAC. Quality Control Procedure. Ref. CMEMS-INS-QualityControlProcedure. Copernicus Marine In Situ TAC. https://doi.org/10.13155/106219.

Pérez-Gómez B, Álvarez-Fanjul E, She J, Pérez-González I, Manzano F. 2016. Extreme sea level events, Section 4.4, p:300. In: Von Schuckmann K, Le Traon PY, Alvarez-Fanjul E, Axell L, Balmaseda M, Breivik LA, Brewin RJW, Bricaud C, Drevillon M, Drillet Y, Dubois C , Embury O, Etienne H, García-Sotillo M, Garric G, Gasparin F, Gutknecht E, Guinehut S, Hernandez F, Juza M, Karlson B, Korres G, Legeais JF, Levier B, Lien VS, Morrow R, Notarstefano G, Parent L, Pascual A, Pérez-Gómez B, Perruche C, Pinardi N, Pisano A, Poulain PM , Pujol IM, Raj RP, Raudsepp U, Roquet H, Samuelsen A, Sathyendranath S, She J, Simoncelli S, Solidoro C, Tinker J, Tintoré J, Viktorsson L, Ablain M, Almroth-Rosell E, Bonaduce A, Clementi E, Cossarini G, Dagneaux Q, Desportes C, Dye S, Fratianni C, Good S, Greiner E, Gourrion J, Hamon M, Holt J, Hyder P, Kennedy J, Manzano-Muñoz F, Melet A, Meyssignac B, Mulet S, Nardelli BB, O'Dea E, Olason E, Paulmier A, Pérez-González I, Reid R, Racault MF, Raitsos DE, Ramos A, Sykes P, Szekely T, Verbrugge N. 2016. The Copernicus Marine Environment Monitoring Service Ocean State Report, Journal of Operational Oceanography. 9 (sup2): 235-320. http://dx.doi.org/10.1080/1755876X.2016.1273446.

Pérez Gómez B., De Alfonso M., Zacharioudaki A., Pérez González I., Álvarez Fanjul E., Müller M., Marcos M., Manzano F., Korres G., Ravdas M., Tamm S. 2018. Sea level, SST and waves: extremes variability. In: Copernicus Marine Service Ocean State Report, Issue 2, Journal of Operational Oceanography, 11:sup1, Chap. 3.1, s79–s88, DOI: https://doi.org/10.1080/1755876X.2018.1489208.

Copernicus Marine in situ TAC (2024). Quality Information Document. In Situ TAC multiparameter products: INSITU_region_PHYBGCWAV_DISCRETE_MYNRT_013_030-036. And Synthetic Quality Overview document (SQO). Ref. CMEMS-INS-QUID-013_030-036, CMEMS-INS-SQO-013_030-036. Copernicus Marine In Situ TAC. https://doi.org/10.13155/75807.

Woodworth, P. L., and D. L. Blackman, 2004: Evidence for Systematic Changes in Extreme High Waters since the Mid-1970s. *J. Climate*, 17, 1190–1197, https://doi.org/10.1175/1520-0442(2004)017<1190:EFSCIE>2.0.CO;2.